

When Adaptation Hurts: Connecting Representational Drift to OOD Failures in MedSAM Fine-Tuning.

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Abstract. Foundation models for medical image segmentation, like prompt-based MedSAM, generalize well across domains and modalities, often in zero or few-shot setups. However, their performance depends on the quality of prompts and the adaptation of the models to custom datasets. This work systematically examines how MedSAM generalizes across diverse medical imaging benchmarks, with six adaptation strategies: full-model and encoder-only LoRA, shallow and deep visual prompt tuning (VPT), and decoder-only and full fine-tuning. Models are trained on the International Skin Imaging Collaboration Challenge (ISIC 2018) dataset and evaluated under clean and increasingly noisy prompts on IN and Out-of-Distribution (OOD) datasets: close-OOD PH2 (dermoscopy), far-OOD BUSI (Breast Ultrasound Images Dataset) and CBIS-DDSM (Curated Breast Imaging Subset of the Digital Database for Screening Mammography). We show that adaptation improves performance on IN and close-OOD data but often reduces performance on far-OOD data. Full fine-tuning provides the best tradeoff, while encoder-only LoRA is the strongest parameter-efficient alternative, outperforming standard LoRA and VPT under far-OOD shifts. Using Centered Kernel Alignment (CKA), we show that far-OOD degradation is strongly associated with drift in decoder representations, whereas encoder similarity alone does not explain robustness. This suggests encoder-only LoRA provides stronger robustness than standard LoRA by adapting the encoder to distribution shift in visual features, while preserving the decoder pathway. We further show that random 0–100 pixel jitter on prompts produces more robust and better performing models. We thus conclude that robust MedSAM adaptation requires the combined consideration of prompt noise exposure, domain shift, and representation preservation. We release our code: <https://github.com/ImSounic/medsam-vpt>.

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1 Introduction

Semantic segmentation is essential in many clinical applications, but requires costly manual annotation. While deep learning can reduce this burden, task-specific models often fail to generalize beyond their training domains. Segmentation foundation models (FMs) such as SAM [23] offer a prompt based alternative. Out-of-the-box SAM performance in medical imaging faces challenges, such as domain shift, weak boundaries, low contrast, and modality-specific appearance changes [17,24]. This has motivated medical segmentation FMs [3,15,16] like MedSAM [15], which was trained on over one million medical images, across 150 tasks, but assumes region of interest (ROI) prompts at inference time.

The first challenge this approach faces is ROI sensitivity. Promptable models that require ROI prompts [15,16] highly depend on the user-provided bounding box, which in clinical settings may be noisy, inconsistent, or unavailable, substantially affecting performance [3,10,26]. Previous work has therefore studied strategies to reduce ROI noise sensitivity, by identifying optimal prompting strategies [10,26] or by introducing perturbed bounding-box prompts [21].

The second challenge is generalization under distribution shift, as models lose performance when applied beyond their training distribution [2,7,19]. Prior work has studied out-of-distribution (OOD) identification [2], robustness under distribution shift [19,23], and generalization across domains [3,17,19].

The third challenge is the adaptation cost. Fine-tuning medical FMs can improve performance, but is computationally expensive. Parameter efficient methods such as LoRA [9] and visual prompt tuning (VPT) [11,27] reduce this cost, but remain sensitive to dataset and prompt quality [10,17,22,26]. As a result, adaptation may improve in-domain performance while degrading robustness to OOD data or noisy ROI prompts. Despite advances, prior work has not systematically connected fine-tuning strategy, prompt robustness, and domain generalization in medical segmentation FMs. Existing studies often focus on target domain adaptation, while separately acknowledging ROI noise and distribution shift. However, it remains unclear which fine-tuning strategies provide the best target-domain performance while preserving generalization, which lose most generalization capabilities, and which are most sensitive to noisy prompts.

Contributions This work studies how fine-tuning strategies affect segmentation performance under prompt and domain shift, and relates performance to the internal representations of MedSAM and its fine-tuned variants. In contrast to prior work that focuses mainly on adaptation, prompt refinement, or prompt automation, we jointly evaluate adaptation strategy, prompt robustness, and domain generalization. We compare zero-shot MedSAM with full and decoder-only fine-tuning, full-model and encoder-only LoRA, and shallow and deep VPT under controlled bounding-box perturbations in training and evaluation.

Our contributions are threefold. First, we systematically evaluate bounding-box perturbation, using models trained with clean boxes, fixed-noise boxes, and variable-noise boxes, then evaluated across multiple perturbation levels. Second, we compare full fine-tuning and parameter-efficient fine-tuning in the same perturbation-aware setting to identify which methods remain robust when ROI prompts are imperfect, while test data shifts away from the target domain. Finally, we link layerwise representational similarity, measured with centered kernel alignment (CKA) [12], to in- and OOD segmentation performance, and test whether representation preservation is associated with OOD generalization.

2 Related Work

Recent progress in foundation models has led to segmentation systems that transfer well to medical imaging. MedSAM [15] is a notable example, adapting SAM [23] to medical images, achieving strong performance across modalities, with extended following versions [15,16,24,25]. Medical datasets are often small and costly to annotate, so foundation models are often evaluated in zero-shot settings. However, out-of-the-box performance is often suboptimal, largely due to distribution shift between pretraining and target clinical data [3,17,23]. Fine-tuning on target domain data can improve performance, but it is computationally expensive, requires annotated data, and may reduce generalization ability.

Therefore, lightweight methods have been explored, like prompt tuning [6], visual prompt tuning [5,8,11,20], and low rank adaptation (LoRA) adapters [6,9]. These methods aim to preserve the benefits of large pre-trained models while reducing training cost, but remain sensitive to task design and prompt quality.

2.1 Efficient fine-tuning methods

MedSAM architecture MedSAM [15] is the backbone to be evaluated under zero-shot and fine-tuned settings. It consists of an image encoder, prompt encoder, and mask decoder. The image encoder is a Vision Transformer (ViT) with 12 transformer blocks that produce dense image embeddings. The prompt encoder embeds user prompts, such as bounding boxes, which the two-layer mask decoder fuses with image embeddings to produce segmentation masks.

Prompt tuning Prompt tuning freezes most of the network and learns only a small number of task specific parameters, reporting good performance in [6].

Visual prompt tuning (VPT) VPT extends prompt adaptation to vision models by learning a set of prompts while freezing the backbone [8,11], improving performance in a low-data regime [27], supported by recent SAM variants [5,20].

Low Rank Adaptation (LoRA) LoRA is a parameter-efficient fine-tuning strategy that updates a model through low-rank trainable weight matrices rather than modifying the full weight tensors, while keeping backbone fixed [6,9].

2.2 Robustness and out-of-distribution (OOD) generalization

Generalization under domain shift remains a challenge in medical segmentation, since models often degrade when transferred across scanners, hospitals, or modalities [2,7]. Related studies have focused on OOD identification [2], robustness under shift [23], and generalization across domains [3,17].

2.3 Prompt sensitivity to noise

Prior work has shown that promptable segmentation models are sensitive to prompt type and quality [3,10,17,24,26], and that perturbing the box can significantly reduce segmentation accuracy, with [3], reporting strong variation across datasets and prompt settings [17,24]. RobustMedSAM [14] claims image encoder preserves medical priors, while the mask decoder governs corruption robustness. Several works address this limitation by refining or augmenting the input prompt [10,26], through bounding box correction [26], refining noisy prompts [10] or by training on perturbed bounding boxes [15,21], where imprecise bounding boxes serve as an unsupervised localization task [22].

2.4 Feature representation comparison

Comparing representations is a well-studied problem in machine learning [12]. Comparing only final embeddings can miss layer-specific changes, originating from fine tuning strategies that adapt different parts of MedSAM. We therefore compare internal feature representations across layers using Centered Kernel Alignment (CKA) [12]. For each model, we compute CKA between its layer representations and those of zero shot MedSAM, using the same input images.

3 Methodology

3.1 Perturbed bounding box prompts

All images are resized to 1024×1024 . The bounding box prompts undergo evaluation perturbations of $p \in \{0, 20, 50, 100, 200\}$, corresponding to mean side length increases of 0%, 1.95%, 4.88%, 9.77%, and 19.53% of image size, and mean box area increases of 0%, 17.61%, 48.85%, 112.67%, and 280.14%, reaching as high as 1327.9% for small lesions in CBIS-DDSM.

During training: We train models under three prompt settings: clean bounding boxes, bounding boxes perturbed by a fixed 20 pixels, and bounding boxes randomly perturbed between 0 and 100 pixels. This allows us to test whether exposure to imperfect prompts improves localization robustness, and whether training with a fixed perturbation causes overfitting to a specific noise level.

During evaluation We evaluate each trained model across five bounding box perturbation levels: 0, 20, 50, 100, and 200 pixels. This tests robustness under increasing prompt noise and evaluates whether models trained with perturbations up to 100 pixels generalize to more severe 200 pixel perturbations in inference.

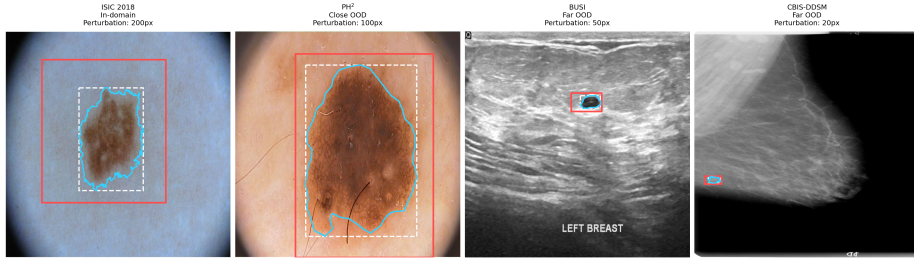


Fig. 1: Prompt perturbation examples.

3.2 Datasets and in- vs out-of-domain separation

We use ISIC 2018 [4] dermoscopic skin lesion images [CC BY-NC] for training and in-domain evaluation. PH² [18] [public research use], another dermoscopic skin lesion dataset, is used as a close out-of-domain dataset because it shares the same imaging modality and target structure but differs in acquisition source and dataset distribution. To evaluate stronger domain shift, we use BUSI breast ultrasound images [1] [CC BY 4.0] and CBIS-DDSM mammography images [13] [TCIA data usage policy] as far out-of-domain datasets. This separation allows testing whether adaptation improves performance only for target domain or preserves generalization across different clinical imaging domains.

3.3 Representation similarity measurement

We measure representation similarity using linear Centered Kernel Alignment (CKA). For each adapted checkpoint, we compare layerwise representations against zero shot MedSAM using the same images and prompts. CKA is computed across the image encoder, prompt encoder, and mask decoder, with higher values indicating stronger preservation of the original MedSAM representation.

4 Experiments and results

4.1 Implementation details

Training paradigms implementation All methods use the MedSAM backbone with a frozen prompt encoder, so bounding-box prompts are encoded identically across experiments. Full fine-tuning updates the image encoder and mask decoder, while decoder-only fine-tuning updates only the mask decoder. Full LoRA adds trainable adapters to both the image encoder and decoder, whereas encoder-only LoRA adds adapters only to the image encoder. VPT freezes the image encoder and adds learnable visual prompt tokens either once at the encoder input for shallow VPT or at every encoder block for deep VPT; in both VPT settings, the mask decoder is trained.

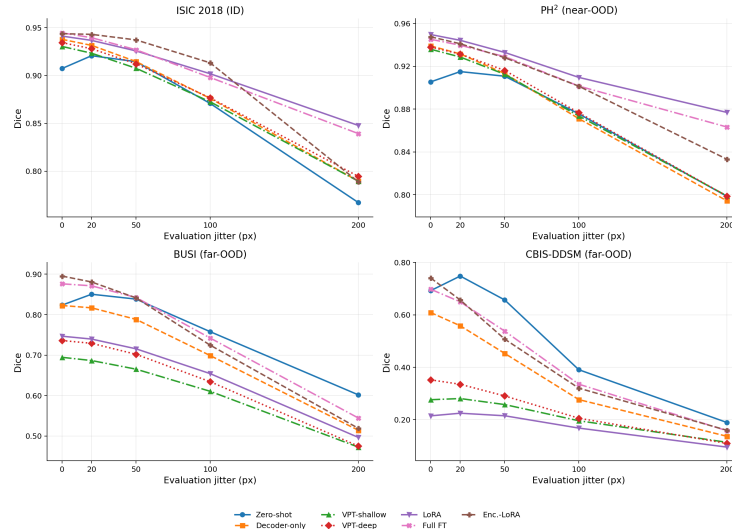


Fig. 2: Robustness to bounding-box perturbation under variable training jitter. Dice is reported across evaluation jitter levels for all adaptation strategies on ID, near-OOD, and far-OOD datasets.

Training details All models were trained for 5 epochs on ISIC 2018 at 1024×1024 resolution using AdamW, mixed precision, and an equally weighted Dice + cross-entropy loss. The 2,594 ISIC images were split 80/10/10 into train, validation, and test sets. Experiments used 8 GPUs with 16–22 GB memory, while training lasted 8 hours. Full fine-tuning used batch size 1 and learning rate 10^{-5} , while parameter-efficient methods used batch size 4 and learning rate 10^{-4} . Hyperparameters were fixed across datasets, Dice was reported for readability, and each configuration was repeated across three random seeds. For LoRA, we used rank ($r=8$), scaling factor ($\alpha = 16$), and dropout (0.0); for VPT, we used 10 prompt tokens per insertion point. The number of trainable parameters was 93.7M for full fine-tuning, 4.4M for full LoRA, 4.1M for encoder-only LoRA, 4.1M for decoder-only fine-tuning, and 4.1M/4.2M for shallow/deep VPT. CKA and OOD evaluation were performed on 200 PH2, 647 BUSI, and 362 CBIS-DDSM images. For ISIC, CKA was computed on all 2,594 images.

4.2 Per-method performance across variable evaluation jitter

Figure 2 summarizes the robustness of each method across jitter levels. For each method, we average performance across the three training prompt perturbations: 0, fixed 20, and random 0–100 pixels. Overall, performance degrades substantially as evaluation jitter increases. The results also show variability across adaptation methods. Full fine-tuning and encoder-only LoRA are the strongest and most robust methods overall. Standard LoRA, drops substantially on OOD

Table 1: Mean Dice gain over zero-shot by regime, method, and training jitter. Values are averaged over three runs; the Mean column reports mean $\pm$ standard deviation. **Bold** and underline indicate the best and second-best methods.

Regime	Method	0	20	r100	Mean	Best train	Best method
ID	Dec. only	-0.016	-0.007	0.014	-0.003 ± 0.002	r100	LoRA
	VPT-S	-0.008	-0.001	0.009	-0.000 ± 0.003	r100	
	VPT-D	-0.010	-0.005	0.013	-0.001 ± 0.002	r100	
	LoRA	0.012	0.016	0.035	0.021 ± 0.001	r100	
	Enc-LoRA	-0.027	-0.028	0.053	-0.001 ± 0.002	r100	
	Full FT	-0.006	-0.005	0.034	0.007 ± 0.002	r100	
Close OOD	Dec. only	-0.016	-0.007	0.009	-0.005 ± 0.004	r100	LoRA
	VPT-S	-0.009	-0.004	0.009	-0.001 ± 0.003	r100	
	VPT-D	-0.011	-0.007	0.011	-0.002 ± 0.001	r100	
	LoRA	0.015	0.020	0.042	0.026 ± 0.001	r100	
	Enc-LoRA	-0.017	-0.022	0.023	-0.005 ± 0.002	r100	
	Full FT	-0.007	-0.003	0.035	0.008 ± 0.002	r100	
Far OOD	Dec. only	-0.036	-0.032	-0.088	-0.052 ± 0.010	20	Full FT
	VPT-S	-0.127	-0.161	-0.230	-0.172 ± 0.002	0	
	VPT-D	-0.126	-0.139	-0.198	-0.154 ± 0.015	0	
	LoRA	-0.142	-0.157	-0.228	-0.176 ± 0.017	0	
	Enc-LoRA	-0.024	-0.052	-0.054	-0.043 ± 0.010	0	
	Full FT	-0.014	-0.011	-0.029	-0.018 ± 0.005	20	
Overall	Dec. only	-0.023	-0.015	-0.022	-0.020 ± 0.005	20	Full FT
	VPT-S	-0.048	-0.055	-0.071	-0.058 ± 0.003	0	
	VPT-D	-0.049	-0.050	-0.058	-0.052 ± 0.004	0	
	LoRA	-0.038	-0.040	-0.051	-0.043 ± 0.006	0	
	Enc-LoRA	-0.023	-0.034	0.007	-0.016 ± 0.004	r100	
	Full FT	-0.009	-0.006	0.013	-0.001 ± 0.003	r100	

datasets, despite strong in-domain performance, as reported in the literature [2]. At higher jitter levels, zero-shot outperforms all adapted models on BUSI and CBIS-DDSM. Only full fine-tuning and encoder-LoRA maintain far OOD performance close to zero-shot while still improving in-domain performance.

4.3 Performance regarding method and training jitter

Table 1 summarizes Dice gain relative to zero-shot MedSAM across adaptation methods and training jitter. Overall, zero-shot performance is difficult to improve upon, especially on far-OOD datasets, while in-domain and close-OOD gains often come with reduced generalization.

Standard LoRA achieves the largest ID and close-OOD gains, but drops sharply under far-OOD shift, with a mean gain of -0.176. In contrast, encoder-only LoRA is more robust, with a far-OOD mean of -0.043 and the second-best overall mean of -0.016. Better generalization by restricting LoRA to the encoder is consistent with the CKA analysis (Section 4.4) that shows preserving decoder representations is associated with far-OOD robustness. Visual prompt tuning does not provide a favorable tradeoff: shallow and deep VPT show negligible ID and close-OOD gains, with large far-OOD drops. Training with random 0–100-pixel jitter generally produces the best adapted models overall, especially for full fine-tuning and encoder-only LoRA.

Across all regimes, full fine-tuning remains the strongest method, achieving the best far-OOD mean gain (-0.018) and best overall mean gain (-0.001).

Table 2: Spearman/Pearson correlations between CKA and Dice gain over zero-shot. Stars denote false discovery rate corrected significance across 30 tests: * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$. Total sample size is $n = 3000$.

CKA feature	ID (ISIC 2018)	Close-OOD (PH ²)	Far-OOD (BUSI/CBIS)
IoU token layer	0.097 / 0.106	-0.018 / -0.021	0.760*** / 0.692***
Output layer	0.060 / 0.026	0.004 / -0.017	0.755*** / 0.718***
Decoder layers	-0.075 / -0.107	-0.022 / -0.079	0.726*** / 0.687***
Upscaled emb.	0.195 / 0.050	0.294* / 0.213	0.713*** / 0.791***
Encoder layers	-0.282* / -0.299*	-0.281* / -0.352**	0.091 / 0.059

Encoder-only LoRA is the strongest parameter-efficient alternative, clearly outperforming standard LoRA and VPT under OOD shift.

4.4 CKA correlation with OOD performance

The following CKA analysis links performance to decoder representational drift, as each training paradigm inevitably shifts internal model representations across layers. Here, we connect which MedSAM internal representations correlate with performance gains. Table 2 shows that CKA is weakly correlated with performance gains in ID and close-OOD regimes, but strongly correlated with far-OOD gains for decoder-side representations. The IoU token output, output layer, decoder layers, and upsampled embedding show high positive far-OOD correlations. In contrast, encoder-layer CKA is weakly negative across regimes, indicating that encoder similarity to zero-shot MedSAM alone does not explain robustness.

Overall, far-OOD degradation is tied to drift in the mask decoder and output representations, suggesting that generalization is harmed when output-side representations are disrupted. Therefore we include encoder-only LoRA fine-tuning, which performs better both overall and on far-OOD data than standard LoRA, while remaining competitive with full FT. This analysis, often ignored by works focusing mainly on Dice optimization, suggests encoder adaptation of MedSAM can result in a much more robust model overall.

Because the pooled CKA observations share images and backbones across methods, seeds, and perturbation levels, they are not independent and the effective sample size is smaller than the nominal n . The reported p -values should therefore be interpreted as descriptive rather than strict hypothesis tests. This does not change the qualitative conclusion that decoder-side similarity tracks far-OOD performance more closely than encoder similarity.

5 Conclusion

We systematically evaluated MedSAM adaptation under joint domain shift and bounding-box prompt perturbation, both during training and evaluation. By training with clean boxes, fixed 20-pixel, and random 0–100 pixel perturbations,

and evaluating across increasing prompt noise, we show that prompt robustness depends strongly on both the adaptation strategy and the training jitter regime. Zero-shot MedSAM remains difficult to improve upon when far-OOD robustness is considered, especially under severe prompt perturbation.

Across adaptation methods, full fine-tuning provides the strongest overall tradeoff, while encoder-only LoRA is the best parameter-efficient alternative. Compared with standard LoRA, decoder-only fine-tuning, and shallow or deep visual prompt tuning, encoder-only LoRA better preserves far-OOD performance while maintaining strong in-domain and close-OOD performance.

Our CKA analysis links performance differences to decoder representational drift. Far-OOD generalization is strongly associated with preserving decoder and output representations, whereas encoder similarity does not explain robustness. This supports encoder-only LoRA as a practical CKA-informed adaptation strategy: adapting the image encoder while avoiding decoder/output drift leads to a stronger robustness tradeoff than standard LoRA. Overall, variable 0–100 pixel jitter training produces the most reliable adapted models.

Disclosure of Interests. The authors have no competing interests in the paper.

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Supplementary Material

A How hard of a task does each evaluation jitter represent?

In Table 3 we report that the same pixel-level jitter corresponds to very different effective prompt difficulty across datasets. For ISIC 2018 and PH², the increase in bounding-box area remains relatively moderate even at larger perturbations, whereas BUSI and especially CBIS-DDSM experience much larger relative ROI expansions, reflecting the smaller lesion sizes and more severe prompt degradation in those domains. This means that a fixed jitter value does not represent the same level of challenge across all datasets, and that far-ODD evaluation is substantially more demanding than in-domain evaluation. Taken together, these results support our use of jitter-aware evaluation as a necessary component for assessing robustness under noisy ROI prompts.

Table 3: Relative increase in tight bounding-box area under expansion due to evaluation jitter. Values report the mean area increase with respect to the original tight bounding box for each dataset and perturbation level.

Dataset	pm=20	pm=50	pm=100	pm=200
ISIC2018-test	11.70%	30.51%	66.40%	152.26%
PH2	6.57%	16.84%	35.11%	72.06%
BUSI	18.71%	51.36%	113.75%	270.96%
CBIS-DDSM	64.06%	193.49%	485.19%	1327.90%
Overall	17.61%	48.85%	112.67%	280.14%

B Boundary robustness under evaluation jitter

Table 4 reports boundary accuracy for clean bounding-box prompts. The adapted models obtain low median HD95 values on ISIC 2018 and PH². On the far-ODD datasets, the differences between methods are larger. Encoder-only LoRA, decoder-only fine-tuning, and full fine-tuning score relatively low boundary errors, while standard LoRA and the VPT variants have much wider interquartile ranges, particularly on BUSI and CBIS-DDSM.

Table 4: HD95 at zero evaluation jitter. Values are median [interquartile range] in pixels at 1024×1024 ; lower is better.

Method	ISIC 2018 (ID)	PH ² (near-OOD)	BUSI (far-OOD)	CBIS-DDSM (far-OOD)
Zero-shot	6 [0, 15]	7 [1, 16]	14 [9, 25]	12 [7, 17]
Dec. only	0 [0, 3]	0 [0, 3]	4 [2, 17]	4 [3, 6]
VPT-S	0 [0, 3]	0 [0, 4]	21 [5, 387]	14 [8, 37]
VPT-D	0 [0, 3]	0 [0, 4]	25 [6, 394]	18 [10, 359]
LoRA	0 [0, 1]	0 [0, 1]	26 [6, 402]	47 [13, 438]
Enc-LoRA	0 [0, 0]	0 [0, 0]	3 [1, 11]	5 [3, 7]
Full FT	0 [0, 0]	0 [0, 0]	4 [1, 13]	4 [3, 6]

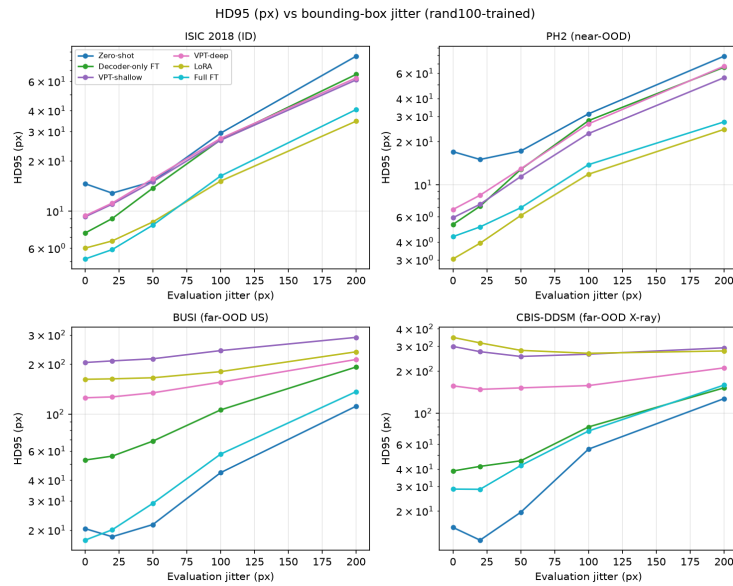


Fig. 3: HD95 across evaluation jitter levels for each adaptation method and dataset. Values are shown on a logarithmic scale in pixels at 1024×1024 ; lower is better.

Figure 3 extends this comparison across evaluation jitter levels. Boundary error generally increases as the bounding box is expanded, as the quality of the prompts degrades, but the rate of degradation differs across adaptation methods and datasets. The largest errors occur for standard LoRA and VPT on the far-OOD datasets.

C How does CKA similarity correspond to segmentation performance

Figures 4 and 5 show that CKA similarity is related to segmentation performance, but the relationship depends strongly on which part of the network is compared. The scatter plots indicate that higher similarity in decoder-side and output-side representations is associated with better far-OOD performance, while encoder similarity alone does not track robustness as reliably. The heatmap makes the same pattern easier to see across datasets and jitter levels, where the strongest correlations appear in the far-OOD regime. Overall, these figures support the claim that preserving decoder and output representations is more important for robustness than preserving encoder features alone.

Figure 4 plots the performance gain over zero-shot against overall CKA similarity for all settings. Each method has its own plot and each point in the scatter plot corresponds to one configuration of the domains (ID, near-OOD, and far-OOD) and training jitter for a given adaptation method. The figure displays how the CKA–performance relationship differs across adaptation strategies while still aggregating over all domain and jitter settings.

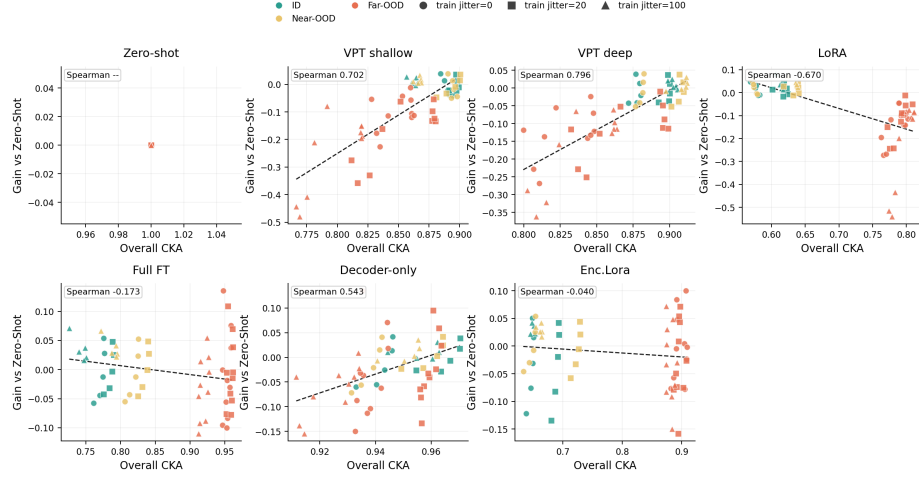


Fig. 4: CKA-performance relationship by method.

Figure 5 shows a plot per regime, where each plot shows the performance gain over zero-shot against overall CKA similarity for all settings. Each point in the scatter corresponds to one configuration of the adaptation method, training jitter, and evaluation jitter. Per method, the plots cover four datasets, three training jitters, and five evaluation jitters per method. The results shown in the four plots show how the relationship changes when moving from in-domain to far-OOD conditions.

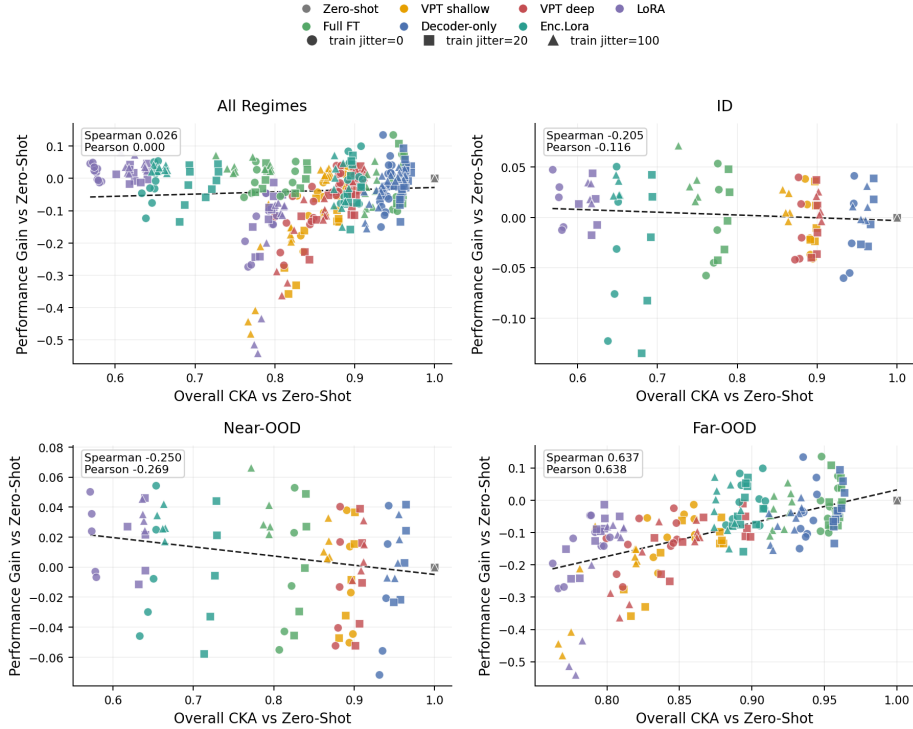


Fig. 5: Overview of the relationship between CKA similarity and segmentation performance.

Figure 6 shows a line plot split into 6 parts. The plot is split into encoder, image emb, prompt, decoder, tokens, and upscale sections. These are the different modules of the model, and the x-axis shows the different components within those modules that we fine-tune. The y-axis shows the linear CKA similarity values. Each line section represents a training method and shows how these different methods alter the internal state of a model compared to its original zero-shot state. The shaded regions around the curves denote variability across datasets and jitters. By showing how fine-tuning affects the similarity scores per method, per component of the model, this plot shows which components are stable and

which are sensitive to the fine-tuning task.

In the encoder (E00–E11), most adaptation strategies largely preserve the representation of the reference MedSAM model. The zero-shot model, by definition, remains at a similarity of 1.0, while adapted variants show modest but method-dependent drift, with approaches such as LoRA and encoder-only LoRA showing the strongest deviations.

In contrast, similarity drops much more sharply in the decoder and token components, particularly for VPT-shallow, VPT deep, and decoder-only tuning, indicating that these parts of the network undergo the largest representational changes during adaptation. The token outputs are substantially reshaped across methods, whereas LoRA variants and full fine-tuning tend to better preserve decoder-side structure, which is consistent with their stronger far-OOD robustness in the main experiments.

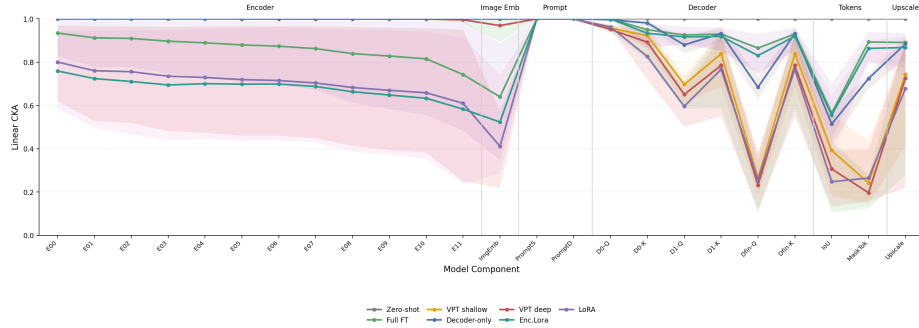


Fig. 6: CKA similarity across datasets and jitters using perfect bounding boxes.

D Degradation of performance depending on bounding box jitter

Figures 7 and 8 show that all methods degrade as bounding-box jitter increases, but they do not degrade at the same rate. Full fine-tuning and encoder-only LoRA have the shallowest degradation curves, especially when they are trained with variable jitter, while standard LoRA and VPT drop more quickly under stronger perturbation. The heatmap makes the same trend visible across datasets, showing that far-OOD performance is much more sensitive to prompt noise than in-domain performance. Specifically for CBIS-DDSM the Dice scores collapse to values below 0.2 for all methods when heavy jitter is introduced to the bounding box dimensions. These results support our claim that exposure to variable prompt noise during training improves robustness under realistic prompt perturbations.

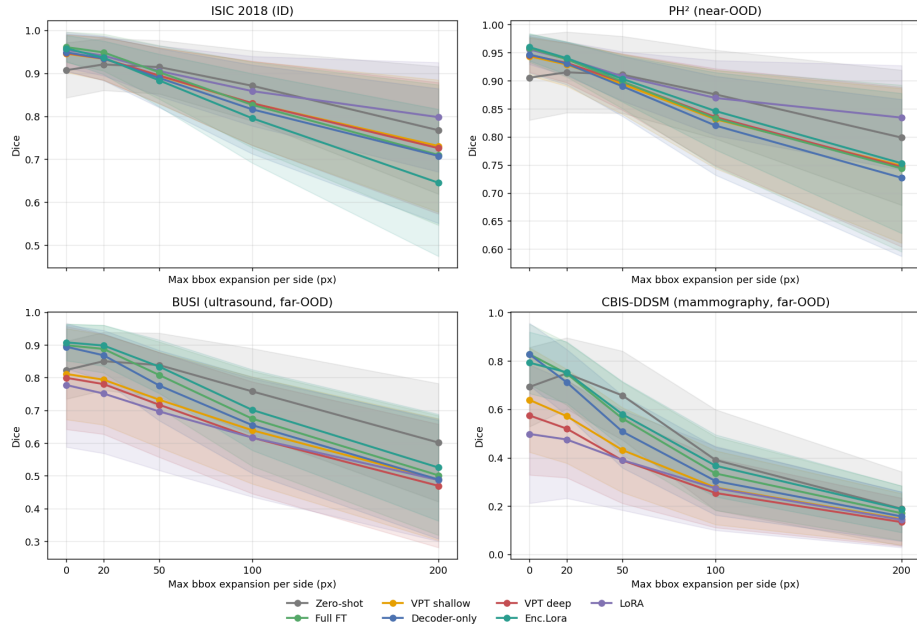


Fig. 7: Dice degradation with increasing bounding-box jitter by dataset and method.

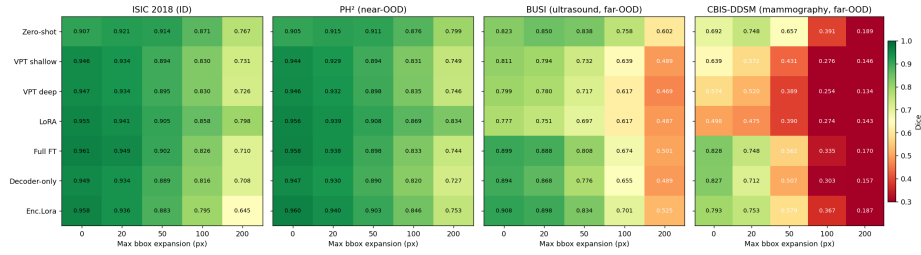


Fig. 8: Heatmap of the degradation in Dice with increasing bounding-box jitter, by datasets across models

E Segmentation performance breakdown across jitter levels

Table 5: Performance and CKA under bounding-box jitter. Dice performance and CKA are reported at zero jitter, averaged across evaluation jitters, and at 200 px jitter. For performance columns, underline marks the best method within each dataset and training-jitter block, while bold marks the best value for each dataset and evaluation column across all training-jitter settings. For CKA columns, bold and underline mark the best and second-best adapted methods, respectively, within each dataset and training-jitter block

Dataset	Training Method	Dice@0	mDice	Dice@200	CKA@0	mCKA	CKA@200
ISIC 2018 <i>ID</i>	0	Zero-shot	0.9072	<u>0.9019</u>	0.7673	1.0000	1.0000
		Decoder-only	0.9487	0.8798	0.7075	0.9462	0.9432
		VPT-shallow	0.9457	0.8858	0.7306	<u>0.8845</u>	<u>0.8908</u>
		VPT-deep	0.9472	0.8863	0.7256	0.8767	0.8791
		LoRA	0.9545	0.9015	<u>0.7978</u>	0.5685	0.5794
		LoRA Encoder-only	0.9580	0.8435	0.6450	0.6484	0.6462
		Full FT	0.9609	0.8924	0.7098	0.7755	0.7738
	20	Zero-shot	0.9072	<u>0.9019</u>	0.7673	1.0000	1.0000
		Decoder-only	0.9462	0.8961	0.7408	0.9703	0.9651
		VPT-shallow	0.9429	0.8948	0.7455	0.8969	0.8965
		VPT-deep	0.9444	0.8890	0.7277	<u>0.8987</u>	<u>0.8981</u>
		LoRA	0.9510	0.8998	<u>0.7808</u>	0.6184	0.6168
		LoRA Encoder-only	0.9496	0.8414	0.6330	0.6925	0.6888
		Full FT	<u>0.9553</u>	0.8987	0.7252	0.7886	0.7851
	rand100	Zero-shot	0.9072	0.9019	0.7673	1.0000	1.0000
		Decoder-only	0.9384	0.9042	0.7782	0.9623	0.9569
		VPT-shallow	0.9316	0.9040	0.7951	0.8645	0.8623
		VPT-deep	0.9330	0.9037	0.7886	<u>0.9017</u>	<u>0.9026</u>
		LoRA	0.9421	0.9240	0.8494	0.6125	0.6136
		LoRA Encoder-only	0.9436	0.9052	0.7890	0.6511	0.6498
		Full FT	<u>0.9446</u>	0.9244	0.8386	0.7488	0.7425
PH ² <i>near-OOD</i>	0	Zero-shot	0.9054	0.9004	0.7986	1.0000	1.0000
		Decoder-only	0.9466	0.8802	0.7269	0.9423	0.9385
		VPT-shallow	0.9436	0.8846	0.7485	<u>0.8905</u>	<u>0.8958</u>
		VPT-deep	0.9458	0.8883	0.7463	0.8825	0.8810
		LoRA	0.9557	<u>0.9053</u>	<u>0.8342</u>	0.5716	0.5766
		LoRA Encoder-only	0.9598	0.8804	0.7528	0.6535	0.6466
		Full FT	0.9583	0.8898	0.7437	0.8258	0.8201
	20	Zero-shot	0.9054	0.9004	0.7986	1.0000	1.0000
		Decoder-only	0.9473	0.9024	0.7753	0.9642	0.9593
		VPT-shallow	0.9421	0.8922	0.7516	0.9004	0.8934
		VPT-deep	0.9444	0.8898	0.7465	<u>0.9067</u>	<u>0.9066</u>
		LoRA	0.9516	<u>0.9032</u>	<u>0.8258</u>	0.6391	0.6337
		LoRA Encoder-only	0.9497	0.8750	0.7408	0.7282	0.7234
		Full FT	<u>0.9543</u>	0.8996	0.7533	0.8398	0.8350
	rand100	Zero-shot	0.9054	0.9004	0.7986	1.0000	1.0000
		Decoder-only	0.9408	0.9053	0.7916	0.9552	0.9512

CBIS-DDSM <i>far-OOD</i>	0	VPT-shallow	0.9382	0.9106	0.8092	0.8674	0.8665	0.8619
		VPT-deep	0.9368	0.9060	0.7899	<u>0.9114</u>	<u>0.9080</u>	<u>0.8975</u>
		LoRA	<u>0.9510</u>	0.9306	0.8724	0.6363	0.6349	0.6241
		LoRA Encoder-only	0.9475	0.9102	0.8332	0.6631	0.6616	0.6544
		Full FT	0.9468	0.9265	0.8651	0.7936	0.7882	0.7718
	20	Zero-shot	0.6924	0.5986	0.1889	1.0000	1.0000	1.0000
		Decoder-only	0.8273	0.5076	0.1570	<u>0.9353</u>	<u>0.9339</u>	<u>0.9345</u>
		VPT-shallow	0.6388	0.4264	0.1461	0.8278	0.8361	0.8531
		VPT-deep	0.5744	0.3876	0.1336	0.7996	0.8104	0.8221
		LoRA	0.4979	0.3795	0.1433	0.7624	0.7710	0.7882
		LoRA Encoder-only	0.7926	0.5358	0.1872	0.9073	0.9070	0.9092
		Full FT	0.8280	0.5480	0.1704	0.9479	0.9487	0.9533
	rand100	Zero-shot	0.6924	0.5986	0.1889	1.0000	1.0000	1.0000
		Decoder-only	0.6527	0.4749	0.1528	<u>0.9115</u>	0.9166	0.9268
		VPT-shallow	0.2496	0.2327	0.1101	0.7664	0.7767	0.7917
		VPT-deep	0.4049	0.3178	0.1241	0.8023	0.8172	0.8358
		LoRA	0.1785	0.2076	0.1073	0.7738	0.7849	0.8003
		LoRA Encoder-only	<u>0.7410</u>	0.4769	0.1592	0.8850	0.8864	0.8920
Full FT		0.7064	0.5159	0.1626	0.9152	<u>0.9137</u>	<u>0.9146</u>	
BUSI <i>far-OOD</i>	0	Zero-shot	0.8234	0.8156	0.6017	1.0000	1.0000	1.0000
		Decoder-only	0.8944	0.7663	0.4888	<u>0.9442</u>	<u>0.9415</u>	<u>0.9370</u>
		VPT-shallow	0.8112	0.7218	0.4889	0.8599	0.8605	0.8625
		VPT-deep	0.7995	0.7047	0.4695	0.8459	0.8466	0.8458
		LoRA	0.7775	0.6882	0.4870	0.7888	0.7952	0.8018
		LoRA Encoder-only	0.9078	0.7732	0.5253	0.8915	0.8887	0.8821
		Full FT	0.8991	0.7901	0.5014	0.9591	0.9568	0.9528
	20	Zero-shot	0.8234	0.8156	0.6017	1.0000	1.0000	1.0000
		Decoder-only	0.8828	0.7963	0.5370	0.9628	<u>0.9602</u>	<u>0.9559</u>
		VPT-shallow	0.7701	0.6944	0.4781	0.8787	0.8787	0.8792
		VPT-deep	0.8136	0.7329	0.4884	0.8935	0.8959	0.8990
		LoRA	0.8109	0.7362	0.5126	0.7981	0.7965	0.7927
		LoRA Encoder-only	<u>0.8949</u>	0.7734	0.5227	0.8972	0.8931	0.8864
		Full FT	0.8935	0.8094	0.5241	<u>0.9611</u>	0.9608	0.9591
	rand100	Zero-shot	0.8234	0.8156	0.6017	1.0000	1.0000	1.0000
		Decoder-only	0.8322	0.7771	0.5114	0.9338	0.9319	0.9291
		VPT-shallow	0.6723	0.6303	0.4509	0.8193	0.8197	0.8198
		VPT-deep	0.7631	0.7070	0.4747	0.8614	0.8620	0.8604
LoRA		0.7474	0.7132	0.5179	0.8039	0.8086	0.8114	
LoRA Encoder-only		<u>0.8954</u>	0.7722	0.5190	0.8740	0.8744	0.8738	

Full FT 0.8783 0.8085 0.5135 0.9247 0.9253 0.9223

Table 6: Dice under bounding-box jitter. Models are trained with 0 px, 20 px, or random 0–100 px jitter and evaluated at 0–200 px jitter. Values are means over 3 seeds; underline denotes the best method within each training block and bold the best per dataset/evaluation jitter.

Dataset	Train	Method	E0	E20	E50	E100	E200
ISIC 2018 <i>trained on</i>	pm=0	Zero-shot	0.9072	0.9206	<u>0.9144</u>	<u>0.8708</u>	0.7673
		Decoder-only	0.9487	0.9349	<u>0.8900</u>	0.8176	0.7102
		VPT-shallow	0.9449	0.9327	0.8952	0.8335	0.7352
		VPT-deep	0.9470	0.9343	0.8944	0.8281	0.7251
		LoRA	0.9556	0.9416	0.9040	0.8522	<u>0.7867</u>
		LoRA Encoder-only	0.9580	0.9364	0.8832	0.7951	0.6450
		Full FT	<u>0.9610</u>	<u>0.9483</u>	0.9024	0.8259	0.7115
	pm=20	Zero-shot	0.9072	0.9206	<u>0.9144</u>	<u>0.8708</u>	0.7673
		Decoder-only	0.9462	0.9365	0.9021	0.8331	0.7271
		VPT-shallow	0.9428	0.9326	0.9042	0.8476	0.7478
		VPT-deep	0.9432	0.9324	0.9014	0.8407	0.7380
		LoRA	0.9492	0.9392	0.9114	0.8629	<u>0.7974</u>
		LoRA Encoder-only	0.9496	0.9411	0.8950	0.7885	0.6330
		Full FT	<u>0.9556</u>	<u>0.9438</u>	0.9050	0.8314	0.7201
	rand100	Zero-shot	0.9072	0.9206	0.9144	0.8708	0.7673
		Decoder-only	0.9377	0.9314	0.9145	0.8761	0.7896
		VPT-shallow	0.9303	0.9232	0.9075	0.8725	0.7896
		VPT-deep	0.9343	0.9279	0.9121	0.8766	0.7946
		LoRA	0.9411	0.9366	0.9256	0.9018	<u>0.8478</u>
		LoRA Encoder-only	0.9436	<u>0.9429</u>	<u>0.9371</u>	<u>0.9132</u>	0.7890
		Full FT	<u>0.9445</u>	0.9394	0.9268	0.8979	0.8393
PH ² <i>near-OOD</i>	pm=0	Zero-shot	0.9054	0.9150	<u>0.9107</u>	<u>0.8756</u>	0.7986
		Decoder-only	0.9467	0.9312	0.8917	0.8229	0.7309
		VPT-shallow	0.9433	0.9298	0.8976	0.8360	0.7542
		VPT-deep	0.9456	0.9315	0.8961	0.8324	0.7425
		LoRA	0.9570	0.9384	0.9045	0.8600	<u>0.8197</u>
		LoRA Encoder-only	0.9598	0.9402	0.9032	0.8459	0.7528
		Full FT	<u>0.9583</u>	<u>0.9385</u>	0.8981	0.8320	0.7434
	pm=20	Zero-shot	0.9054	0.9150	0.9107	<u>0.8756</u>	0.7986
		Decoder-only	0.9449	0.9329	0.9019	0.8382	0.7513
		VPT-shallow	0.9426	0.9310	0.9040	0.8482	0.7591
		VPT-deep	0.9428	0.9308	0.9022	0.8431	0.7533
		LoRA	0.9511	0.9379	<u>0.9128</u>	0.8717	<u>0.8338</u>
		LoRA Encoder-only	0.9497	0.9363	0.9051	0.8429	0.7408
		Full FT	<u>0.9537</u>	<u>0.9398</u>	0.9044	0.8400	0.7516
	rand100	Zero-shot	0.9054	0.9150	0.9107	0.8756	0.7986
		Decoder-only	0.9390	0.9315	0.9132	0.8711	0.7944

Dataset	Train	Method	E0	E20	E50	E100	E200
BUSI <i>far-OOD</i>		VPT-shallow	0.9359	0.9286	0.9125	0.8736	0.7990
		VPT-deep	0.9379	0.9314	0.9161	0.8766	0.7988
		LoRA	<u>0.9496</u>	0.9442	0.9329	0.9096	0.8769
		LoRA Encoder-only	0.9475	0.9410	0.9280	0.9014	0.8332
		Full FT	0.9452	0.9396	0.9290	0.9015	0.8633
	pm=0	Zero-shot	0.8234	0.8504	0.8384	0.7579	0.6017
		Decoder-only	0.8935	0.8693	0.7785	0.6587	0.4915
		VPT-shallow	0.7921	0.7758	0.7190	0.6315	0.4868
		VPT-deep	0.8017	0.7842	0.7243	0.6288	0.4824
		LoRA	0.7801	0.7540	0.6975	0.6186	0.4903
		LoRA Encoder-only	0.9078	0.8983	0.8336	0.7012	0.5253
		Full FT	0.9007	0.8860	0.8051	0.6722	0.4996
	pm=20	Zero-shot	0.8234	0.8504	0.8384	<u>0.7579</u>	<u>0.6017</u>
		Decoder-only	0.8807	0.8717	0.8088	0.6879	0.5184
		VPT-shallow	0.7623	0.7509	0.7091	0.6261	0.4790
		VPT-deep	0.7702	0.7624	0.7224	0.6360	0.4853
		LoRA	0.7745	0.7586	0.7162	0.6347	0.5010
		LoRA Encoder-only	0.8949	<u>0.8937</u>	0.8470	0.7089	0.5227
		Full FT	<u>0.8996</u>	0.8889	0.8236	0.6902	0.5096
	rand100	Zero-shot	0.8234	0.8504	0.8384	<u>0.7579</u>	<u>0.6017</u>
		Decoder-only	0.8226	0.8168	0.7883	0.6989	0.5142
		VPT-shallow	0.6945	0.6865	0.6652	0.6105	0.4725
		VPT-deep	0.7360	0.7291	0.7019	0.6343	0.4751
		LoRA	0.7467	0.7396	0.7155	0.6542	0.4965
		LoRA Encoder-only	<u>0.8954</u>	<u>0.8807</u>	0.8414	0.7246	0.5190
		Full FT	0.8762	0.8712	<u>0.8424</u>	0.7422	0.5445
CBIS-DDSM <i>far-OOD</i>	pm=0	Zero-shot	0.6924	0.7483	0.6570	0.3905	0.1889
		Decoder-only	0.8280	0.7059	0.5039	0.3026	0.1574
		VPT-shallow	0.5662	0.5199	0.3961	0.2565	0.1374
		VPT-deep	0.5725	0.5102	0.3904	0.2582	0.1368
		LoRA	0.5094	0.4782	0.3883	0.2703	0.1467
		LoRA Encoder-only	0.7926	0.7530	0.5792	0.3668	0.1872
		Full FT	0.8285	0.7438	0.5618	0.3364	0.1713
	pm=20	Zero-shot	0.6924	<u>0.7483</u>	<u>0.6570</u>	<u>0.3905</u>	<u>0.1889</u>
		Decoder-only	0.7824	0.7012	0.5186	0.3040	0.1532
		VPT-shallow	0.4555	0.4369	0.3532	0.2380	0.1313
		VPT-deep	0.5117	0.4810	0.3910	0.2603	0.1369
		LoRA	0.4370	0.4193	0.3580	0.2472	0.1327
		LoRA Encoder-only	0.7467	0.6743	0.4987	0.3153	0.1628
		Full FT	<u>0.8051</u>	0.7398	0.5705	0.3414	0.1714
	rand100	Zero-shot	0.6924	<u>0.7483</u>	<u>0.6570</u>	<u>0.3905</u>	<u>0.1889</u>
		Decoder-only	0.6090	0.5583	0.4528	0.2764	0.1362
		VPT-shallow	0.2759	0.2804	0.2574	0.1956	0.1132
		VPT-deep	0.3518	0.3349	0.2903	0.2053	0.1091
		LoRA	0.2143	0.2244	0.2146	0.1676	0.0949

Dataset	Train	Method	E0	E20	E50	E100	E200
		LoRA Encoder-only	<u>0.7410</u>	0.6569	0.5074	0.3201	0.1592
		Full FT	0.6980	0.6497	0.5380	0.3353	0.1575